1 The curve $y = \mathrm{f}(x)$ passes through the point (0, 69) and has gradient given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{3}y - 15\right)^{\frac{1}{3}}.$$

(i) Find $\mathrm{f}(x)$. [4]

(ii) Find the coordinates of the point on the curve where the gradient is 4. [2]

Qn	Solution	Comments
1i	$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{3}y - 15\right)^{\frac{1}{3}}$ $\left(\frac{1}{3}y - 15\right)^{-\frac{1}{3}} \frac{\mathrm{d}y}{\mathrm{d}x} = 1$ $\int \left(\frac{1}{3}y - 15\right)^{-\frac{1}{3}} \mathrm{d}y = \int 1 \ \mathrm{d}x$ $\frac{\left(\frac{1}{3}y - 15\right)^{\frac{2}{3}}}{\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)} = x + C$ $\frac{9}{2}\left(\frac{1}{3}y - 15\right)^{\frac{2}{3}} = x + C$ When $x = 0$, $y = 69$, $C = \frac{9}{2}(23 - 15)^{\frac{2}{3}}$ $= 18$ $\frac{9}{2}\left(\frac{1}{3}y - 15\right)^{\frac{2}{3}} = x + 18$ $\left(\frac{1}{3}y - 15\right)^{\frac{2}{3}} = \frac{2}{9}x + 4$ $\frac{1}{3}y - 15 = \left(\frac{2}{9}x + 4\right)^{\frac{3}{2}}$ $y = 3\left(\left(\frac{2}{9}x + 4\right)^{\frac{3}{2}} + 15\right)$ $\therefore \mathrm{f}(x) = 3\left(\left(\frac{2}{9}x + 4\right)^{\frac{3}{2}} + 15\right)$	We need to make y the subject because we are finding f(x) which is y.

Qn	Solution	Comments
ii	$\frac{\mathrm{d}y}{\mathrm{d}x}=4$ $\left(\frac{1}{3}y-15\right)^{\frac{1}{3}}=4$ $y=237$ $\frac{9}{2}\left(\frac{1}{3}(237)-15\right)^{\frac{2}{3}}=x+18$ $x=54$ $(54,237)$ is the point on the curve with gradient 4.	

2 **(a)** One of the roots of the equation $4x^4-20x^3+sx^2-56x+t=0$, where s and t are real, is $2-3\mathrm{i}$.
Find the other roots of the equation and the values of s and t. [5]

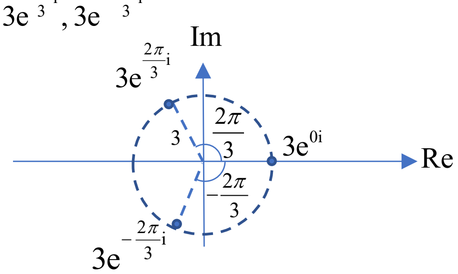
(b) The complex number w is such that $w^3=27$.

(i) Given that one possible value of w is 3, use a **non-calculator method** to find the other possible values of w. Give your answers in the form $a+\mathrm{i}b$, where a and b are exact values. [3]

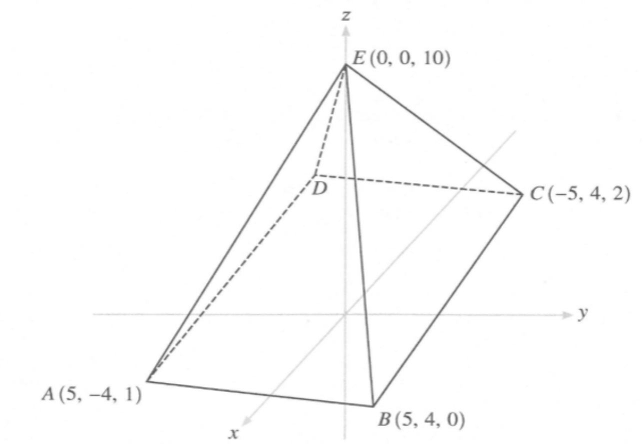
(ii) Write these values of w in modulus-argument form and represent them on an Argand diagram. [2]

(iii) Find the sum and the product of all the possible values of w, simplifying your answers. [2]

Qn	Solution	Comments
2a	Since 2 – 3i is a root, $4(2-3\mathrm{i})^4-20(2-3\mathrm{i})^3+s(2-3\mathrm{i})^2-56(2-3\mathrm{i})+t=0$ $4(-119+120\mathrm{i})-20(-46-9\mathrm{i})+s(-5-12\mathrm{i})-56(2-3\mathrm{i})+t=0$ $332-5s+t+\mathrm{i}(828-12s)=0$ Comparing real and imaginary parts, $828-12s=0\Rightarrow s=69$ $332-5(69)+t=0\Rightarrow t=13$ $\therefore 4x^4-20x^3+69x^2-56x+13=0$ Since all coefficients are real, 2 + 3i is also a root. From GC, another root is $x=\frac{1}{2}$. $\therefore$ The other roots of the equation are 2 + 3i and $\frac{1}{2}$. <u>Alternative</u> Since all coefficients are real, 2 – 3i is a root $\Rightarrow$ 2 + 3i is also a root. Therefore,	We can use calculators to evaluate powers of 2 – 3i. Using GC to solve $4x^4-20x^3+69x^2-56x+13=$ will give the other root as $0.5\pm 2.058\times10^{-7}\mathrm{i}$ When we encounter terms like 2.058×10^{-7}, we should suspect that this could be actually 0, but GC didn't manage to calculate

Qn	Solution	Comments
	$(x-(2-3\mathrm{i}))(x-(2+3\mathrm{i}))=[(x-2)+3\mathrm{i}][(x-2)-3\mathrm{i}]$ $=(x-2)^2+9$ $=x^2-4x+13$ is a quadratic factor. $4x^4-20x^3+sx^2-56x+t=(x^2-4x+13)(px^2-qx+r)$ Comparing coefficient of $x^4: p=4$ Comparing coefficient of x^3: $-4p-q=-20\Rightarrow q=20-4p=4$ Comparing coefficient of x: $-4r-13q=-56\Rightarrow r=\frac{56-13q}{4}=1$ Comparing coefficient of x^2: $13p+r+4q=s\Rightarrow s=69$ Comparing coefficient of x^0: $t=13r=13$ $\Rightarrow$ s = 16 + 52 +1 = 69, t = 13. $\therefore 4x^4-20x^3+69x^2-56x+13=0$ From GC, another root is $x=\frac{1}{2}$. $\therefore$ The other roots of the equation are 2 + 3i and $\frac{1}{2}$.	accurate. We can manually check that $x=\frac{1}{2}$ (repeated root) satisfy the equation.
bi	$w^3-27=(w-3)(w^2+aw+9)$ Compare coefficient of w $-3a+9=0$ $a=3$ $w^2+3w+9=0$ $w=\frac{-3\pm\sqrt{9-36}}{2}$ $=-\frac{3}{2}\pm\frac{3\sqrt{3}}{2}\mathrm{i}$	The question said "non-calculator method". Read question please.
ii	$w=3\mathrm{e}^{0\mathrm{i}}, 3\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}, 3\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$ 	The dotted circle is to show that all the points are the same distance from the origin. We should indicate the modulus and argument on the diagram.
iii	Sum of all roots = 3 + $\left(-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i}\right)+\left(-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i}\right)=0$ Product of all roots = $3\mathrm{e}^{0\mathrm{i}}(3\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}})(3\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}})=27\mathrm{e}^{0\mathrm{i}}=27$	Use cartesian form for sum and polar form for product. Recall: addition and subtraction – Cartesian form faster. Multiply, divide, take power, polar form faster.

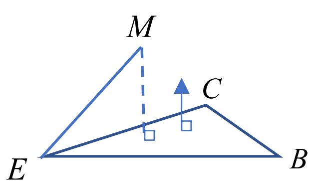
3



An oblique pyramid has a plane base *ABCD* in the shape of a parallelogram. The coordinates of *A*, *B* and *C* are (5, –4, 1), (5, 4, 0) and (–5, 4, 2) respectively. The apex of the pyramid is at $E(0, 0, 10)$ (see diagram).

(i) Find the coordinates of *D*. [1]
(ii) Find the Cartesian equation of face *BCE*. [3]
(iii) Find the angle between face *BCE* and the base of the pyramid. [3]
(iv) Find the shortest distance from the midpoint of edge *AD* to face *BCE*. [5]

Qn	Solution	Comments
3i	*ABCD* is a parallogram, so $\overrightarrow{AD} = \overrightarrow{BC}$ $\overrightarrow{OD} - \begin{pmatrix} 5 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \\ 2 \end{pmatrix}$ $\overrightarrow{OD} = \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix}$ The coordinates of *D* are (–5, –4, 3).	
ii	Method 1 $\overrightarrow{BE} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -4 \\ 10 \end{pmatrix}$ Normal vector of *BCE* = $\begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} -5 \\ -4 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix} = 200$ Cartesian equation of *BCE* is 4*x* + 45*y* + 20*z* = 200. Method 2 Let the cartesian equation be $ax + by + cz = d$	$\begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix}$ is actually $\frac{1}{2}\begin{pmatrix} -10 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{2}\overrightarrow{BC}$. We could leave it as $\overrightarrow{BC}$ if we wanted, but doing so will make calculations slightly easier. Remember to write in Cartesian form.

Qn	Solution	Comments
	(5,4,0), (-5,4,2) and (0,0,10) are points on the plane, $5a+4b=d$ $-5a+4b+2c=d$ $10c=d$ From GC, $a=\frac{1}{50}d$ $b=\frac{9}{40}d$ $c=\frac{1}{10}d$ $\Rightarrow \frac{1}{50}dx+\frac{9}{40}dy+\frac{1}{10}dz=d$ $\therefore 4x+45y+20z=200$	
iii	$\overrightarrow{AB}=\begin{pmatrix}5\\4\\0\end{pmatrix}-\begin{pmatrix}5\\-4\\1\end{pmatrix}=\begin{pmatrix}0\\8\\-1\end{pmatrix}$ Normal vector of $ABCD=\begin{pmatrix}0\\8\\-1\end{pmatrix}\times\begin{pmatrix}-5\\0\\1\end{pmatrix}=\begin{pmatrix}8\\5\\40\end{pmatrix}$ Let θ be the angle between BCE and base of pyramid. $\cos\theta=\frac{\begin{pmatrix}4\\45\\20\end{pmatrix}\cdot\begin{pmatrix}8\\5\\40\end{pmatrix}}{\sqrt{2441}\sqrt{1689}}=\frac{1057}{\sqrt{2441}\sqrt{1689}}$ $\theta=58.6°$ Acute angle between BCE and the base is $58.6°$.	Face BCE is a plane and base of pyramid is also a plane. This question is about **finding angle between two planes**. Recall angle between 2 planes is the angle between the 2 normal vectors. Question: What if the angle between the 2 normals that you found happens to be an obtuse angle?
iv	Let M be midpoint of AD. $\overrightarrow{OM}=\frac{\begin{pmatrix}5\\-4\\1\end{pmatrix}+\begin{pmatrix}-5\\-4\\3\end{pmatrix}}{2}=\begin{pmatrix}0\\-4\\2\end{pmatrix}$  $\overrightarrow{ME}=\begin{pmatrix}0\\0\\10\end{pmatrix}-\begin{pmatrix}0\\-4\\2\end{pmatrix}=\begin{pmatrix}0\\4\\8\end{pmatrix}$ Shortest distance $=\left\|\begin{pmatrix}0\\4\\8\end{pmatrix}\cdot\begin{pmatrix}4\\45\\20\end{pmatrix}\frac{1}{\sqrt{2441}}\right\|=\frac{340}{\sqrt{2441}}$ (or 6.88) Shortest distance from midpoint of AD to BCE is 6.88 units.	This question is about find the distance between a point to a plane. The most common method is to consider the length of projection of ME onto the normal vector of plane BCE.

4 In this question you may use expansions from the List of Formulae (MF26).

(i) Find the Maclaurin expansion of $\ln(\cos 2x)$ in ascending powers of x, up to and including the term in x^6. State any value(s) of x in the domain $0 \leqslant x \leqslant \frac{\pi}{4}$ for which the expansion is **not** valid. [6]

(ii) Use your expansion in part **(i)** and integration to find an approximate expression for $\int \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x$. Hence find an approximate value for $\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x$, giving your answer to 4 decimal places. [3]

(iii) Use your graphing calculator to find a second approximate value for $\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x$, giving your answer to 4 decimal places. [1]

Qn	Solution	Comments
4i	$\ln(\cos 2x) = \ln\left(1 - \frac{4x^2}{2} + \frac{16x^4}{4!} - \frac{64x^6}{6!} + \dots\right)$ $= \ln\left(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + \dots\right)$ $\approx -2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 - \frac{1}{2}\left(-2x^2 + \frac{2}{3}x^4\right)^2 + \frac{1}{3}\left(-2x^2\right)^3$ $= -2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 - \frac{1}{2}\left(4x^4 - \frac{8}{3}x^6\right) + \frac{1}{3}\left(-8x^6\right)$ $= -2x^2 - \frac{4}{3}x^4 - \frac{64}{45}x^6$ $\ln(x)$ is not defined when $x = 0$, so The expansion of $\ln(\cos 2x)$ is not valid when $\cos 2x = 0$. Hence, the expansion is not valid when $x = \frac{\pi}{4}$.	In this question, we should use standard expansions, instead of differentiating (why? How many times do we need to differentiate to get x^6 term?).
ii	$\int \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x \approx \int -2 - \frac{4}{3}x^2 - \frac{64}{45}x^4\,\mathrm{d}x$ $= -2x - \frac{4}{9}x^3 - \frac{64}{225}x^5 + C$ $\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x = \left[-2x - \frac{4}{9}x^3 - \frac{64}{225}x^5\right]_0^{0.5}$ $= -2(0.5) - \frac{4}{9}(0.5)^3 - \frac{64}{225}(0.5)^5$ $= -1.0644\,(4\text{dp})$	
iii	$\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2}\,\mathrm{d}x = -1.0670\,(4\text{dp})$	Remember to use GC to evaluate in radian mode.

5 The manufacturer of a certain type of fan used for cooling electronic devices claims that the mean time to failure (MTTF) is 65 000 hours. The quality control manager suspects that the MTTF is actually less than 65 000 hours and decide to carry out a hypothesis test on a sample of these fans. (An accelerated testing procedure is used to find the MTTF.)

(i) Explain why the manager should take a sample of at least 30 fans, and state how these fans should be chosen. [2]

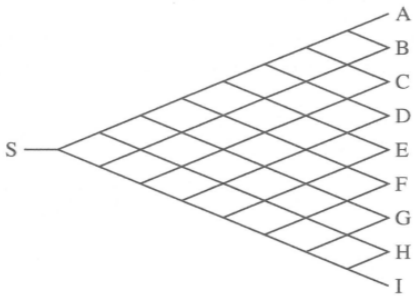
(ii) State suitable hypotheses for the test, defining any symbols that you use. [2]

The quality control manager takes a suitable sample of 43 fans, and finds that they have an MTTF of 64 230 hours.

(iii) Given that the manager concludes that there is no reason to reject the null hypothesis at the 5% level of significance, find the range of possible values of the variance used in calculating the test statistic. [3]

Qn	Solution	Comments
5i	The manager may not know the distribution of Time-To-Failure (TTF). The manager should take a sample of at least 30 fans so that the sample Mean-Time-To-Failure (MTTF) can be approximated by a normal distribution using the Central Limit Theorem. The fans chosen should form a random sample, that is, all the fans should have equal probability of being chosen and the fans are chosen independently.	Why need 30? The trick is to see that this is a hypothesis testing question and that we need CLT to approximate MTTF by a normal distribution to proceed.
ii	Let X hours be the TTF of a fan and let μ hours be the population MTTF of the fans, i.e. $E(X)=\mu$. H_0: $\mu = 65000$ H_1: $\mu < 65000$	
iii	Level of significance: 5% Test statistic: Since n = 43 is large, by CLT, $\overline{X}$ is approximately normal. When H_0 is true, $Z=\dfrac{\overline{X}-65000}{\frac{S}{\sqrt{n}}} \sim \mathrm{N}(0,1)$ approximately. Rejection region: $z \le -1.6449$ Since H_0 is not rejected, $\dfrac{64230-65000}{\frac{s}{\sqrt{43}}} > -1.6449$ $\dfrac{s}{\sqrt{43}} > 468.11$ $s > 3069.6$ $s^2 > 9420000 \left(3\text{sf}\right)$	Use unbiased estimate of population variance since there is no mention of that population is known. 

6



In a computer game, a bug moves from left to right through a network of connected paths. The bug starts at S and, at each junction, randomly takes the left fork with probability p or the right fork with probability q, where $q = 1 - p$. The forks taken at each junction are independent. The bug finishes its journey at one of the 9 endpoints labelled A – I (see diagram).

(i) Show that the probability that the bug finishes its journey at D is $56p^5q^3$. [2]

(ii) Given that the probability that the bug finishes its journey at D is greater than the probability that the bug finishes its journey at any one of the other endpoints, find exactly the possible range of values of p. [4]

In another version of the game, the probability that, at each junction, the bug takes the left fork is $0.9p$, the probability that the bug takes the right fork is $0.9q$ and the probability that the bug is swallowed up by a 'black hole' is 0.1.

(iii) Find the probability that, in this version of the game, the bug reaches one of the endpoints A – I, without being swallowed up by a black hold. [1]

Qn	Solution	Comments
6i	The bug has to choose at a fork 8 times. In order to arrive at D, the bug has to choose left fork 5 times, and the right fork 3 times, in any order. Method 1: Let X be the number of left forks the bug took, out of 8. $X \sim \mathrm{B}(8, p)$ $\mathrm{P}(X = 5) = {}^8C_5\, p^5(1-p)^3 = 56\, p^5q^3$ Method 2: There is $\dfrac{8!}{5!3!}$ number of ways to take 5 left forks and 3 right forks. P(bug arrives at D) = $\dfrac{8!}{5!3!}p^5q^3 = 56p^5q^3$	"Show" question, we are expected to explain clearly. We can list a few possible routes from S to D and try to observe a pattern in the routes. For Method 1, check that the conditions for binomial distribution are satisfied $\Rightarrow$ 8 'trials' – the forks, 'success' when he choose the right fork. Method 2 is like a probability tree method. p^5q^3 is the probability of 5 lefts, followed by 3 rights. We need to find out how many such branches are there, i.e need to arrange 5 L's, 3 R's.
ii	Continuing from method 1 in (i), for the probability to reach D to be the greatest, $X = 5$ is the mode.	It is easier to do this, if you recognise a binomial distribution. Landing at D correspond to 3 right forks and other letters

Qn	Solution	Comments
	$\text{P}(X=5)>\text{P}(X=4)$ and $\text{P}(X=5)>\text{P}(X=6)$ $56p^5(1-p)^3>70p^4(1-p)^4$ and $56p^5(1-p)^3>28p^6(1-p)^2$ $4p>5(1-p)$ and $2(1-p)>p$ $p>\frac{5}{9}$ and $p<\frac{2}{3}$ $\therefore \frac{5}{9}<p<\frac{2}{3}$	correspond to other number of right forks. Recall that for a binomial distribution, there is only 1 peak. Hence, we only need to compare the probability of X = 5 with its neighbours, i.e X = 4 and X = 6.
iii	Method 1: P(bug is not swallowed at a junction) = 1 – 0.1 = 0.9 P(bug reaches exit) = $(0.9)^8$ = 0.430 Method 2: Let Y be no. of junctions where the bug is not swallowed up by a black hole. $Y\sim\text{B}(8, 0.9)$ P(Y = 8) = 0.430	While there appears to be 3 outcomes, we are more concerned with whether the bug is swallowed by black hole, or not.

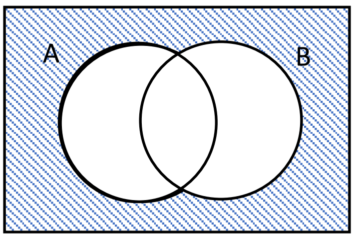
7 The events A, B and C are such that $\text{P}(A)=a$, $\text{P}(B)=b$ and $\text{P}(C)=c$. A and B are independent events. A and C are mutually exclusive events.

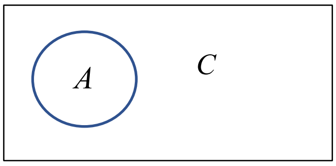
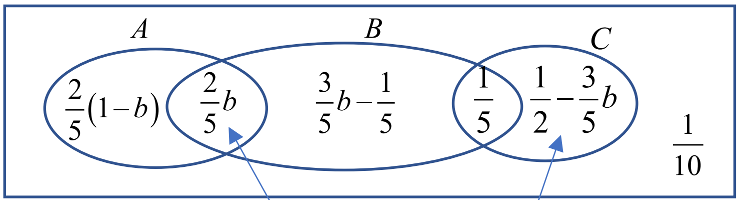
(i) Find an expression for $\text{P}(A'\cap B')$ and hence prove that A' and B' are independent events. [2]

(ii) Find an expression for $\text{P}(A'\cap C')$. Draw a Venn diagram to illustrate the case when A' and C' are also mutually exclusive events. (You should not show event B on your diagram.) [2]

You are now given that A' and C' are **not** mutually exclusive, $P(A)=\frac{2}{5}$, $\text{P}(B\cap C)=\frac{1}{5}$ and $\text{P}(A'\cap B'\cap C')=\frac{1}{10}$.

(iii) Find exactly the maximum and minimum possible values of $\text{P}(A\cap B)$. [4]

Qn	Solution	Comments
7i	$\text{P}(A\cup B)=\text{P}(A)+\text{P}(B)-\text{P}(A\cap B)$ $=a+b-ab$ $\text{P}(A'\cap B')=1-\text{P}(A\cup B)$ $=1-(a+b-ab)$ $=1-a-b+ab$ $=1-a-b(1-a)$ $=(1-a)(1-b)$ $=\text{P}(A')\text{P}(B')$  Since $\text{P}(A'\cap B')=\text{P}(A')\text{P}(B')$, A' and B' are independent events.	This formula is generic. It works for **all** situations. To prove independence between A' and C', we need to show $\text{P}(A'\cap B')=\text{P}(A')\text{P}(B')$

Qn	Solution	Comments
ii	A and C are mutually exclusive $\Rightarrow\ \mathrm{P}(A\cup C)=\mathrm{P}(A)+\mathrm{P}(C)=a+c$ $\mathrm{P}(A'\cap C')=1-\mathrm{P}(A\cup C)=1-(a+c)$ For A' and C' to be mutually exclusive, $\mathrm{P}(A'\cap C')=0$, i.e. $a+c=1$. 	Using same idea in part (i)'s venn diagram. The venn diagram here shows that A' = C. Additional comments for probability: A and B independent $\Rightarrow$ A' and B' independent A and C mutually exclusive $\not\Rightarrow$ A' and C' mutually exclusive
iii	 $P(A\cap B)=P(A)P(B)=\frac{2}{5}b$ $\mathrm{P}(A'\cap B')=\left(1-\frac{2}{5}\right)(1-b)=\frac{3}{5}-\frac{3}{5}b$ $\mathrm{P}(A'\cap B'\cap C)=\frac{3}{5}-\frac{3}{5}b-\frac{1}{10}=\frac{1}{2}-\frac{3}{5}b$ Since $\frac{1}{2}-\frac{3}{5}b\geq 0\Rightarrow b\leq\frac{5}{6}$ $\mathrm{P}(A\cap B)\leq\frac{2}{5}\left(\frac{5}{6}\right)=\frac{1}{3}$ Since $\frac{3}{5}b-\frac{1}{5}\geq 0\Rightarrow b\geq\frac{1}{3}$ $\mathrm{P}(A\cap B)\geq\frac{2}{5}\left(\frac{1}{3}\right)=\frac{2}{15}$ Hence, maximum and minimum $\mathrm{P}(A\cap B)$ are $\frac{1}{3}$ and $\frac{2}{15}$ respectively.	For this part, need to recall that the conditions given in the first paragraph and **the result in (i) will still hold.** Draw a big Venn Diagram and fill in the probabilities of each of the individual regions in the Venn Diagram. For max value, we want $\frac{2}{5}b$ to be as large as possible, so b has to be as large as possible. BUT $0\leq$ probabilities of each of the region ≤ 1 is the restriction that we need to deal with. So it happens that b is max at most $\frac{5}{6}$. For min value, we need b to be as small as possible, but the smallest it can go is $\frac{1}{3}$.

8 A bag contains $(n+5)$ numbered balls. Two of the balls are numbered 3, three of the balls are numbered 4 and n of the balls are numbered 5. Two balls are taken, at random and without replacement, from the bag. The random variable S is the sum of the numbers on the two balls taken.

(i) Determine the probability distribution of S. [4]

(ii) For the case where $n=1$, find $\mathrm{P}(S=10)$ and explain this result. [1]

(iii) Show that $\mathrm{E}(S)=\dfrac{10n+36}{n+5}$ and $\mathrm{Var}(S)=\dfrac{\mathrm{g}(n)}{(n+5)^2(n+4)}$ where $\mathrm{g}(n)$ is a quadratic polynomial to be determined. [6]

<table>
<tr><th>Qn</th><th>Solution</th><th>Comments</th></tr>
<tr><td>8i</td><td>
<table>
<tr><th>s</th><th>$\mathrm{P}(S=s)$</th></tr>
<tr><td>6</td><td>$\frac{2}{(n+5)}\frac{1}{(n+4)}==\frac{2}{(n+5)(n+4)}$</td></tr>
<tr><td>7</td><td>$\frac{2}{(n+5)}\frac{3}{(n+4)}\times 2==\frac{12}{(n+5)(n+4)}$</td></tr>
<tr><td>8</td><td>$\frac{3}{(n+5)}\frac{2}{(n+4)}+\frac{2}{(n+5)}\frac{n}{(n+4)}\times 2=\frac{6+4n}{(n+5)(n+4)}$</td></tr>
<tr><td>9</td><td>$\frac{3}{(n+5)}\frac{n}{(n+4)}\times 2=\frac{6n}{(n+5)(n+4)}$</td></tr>
<tr><td>10</td><td>$\frac{n}{(n+5)}\frac{n-1}{(n+4)}=\frac{n(n-1)}{(n+5)(n+4)}$</td></tr>
</table>
</td><td>Recall 2019 VJC Prelim Q9. Some resemblance.

Good habit to check that the sum of the probabilities is equal to 1.

We can also use the ${}^nC_r$ method, you can get the formula ${}^nC_r=\dfrac{n!}{r!(n-r)!}$ from MF26. E.g
$\mathrm{P}(S=6)$
$=\dfrac{{}^2C_2}{{}^{(n+5)}C_2}$
$=\dfrac{1}{\frac{(n+5)!}{2!(n+5-2)!}}$
$=\dfrac{2(n+3)!}{(n+5)!}$
$=\dfrac{2(n+3)!}{(n+3)!(n+4)(n+5)}$
$=\dfrac{2}{(n+4)(n+5)}$</td></tr>
<tr><td>ii</td><td>When $n=1$, $\mathrm{P}(S=10)=0$
$n=1$ means only 1 ball numbered “5”. To get the sum of 2 numbers on the 2 balls to be equal to 10, we need at least 2 balls numbered 5.</td><td></td></tr>
</table>

<table>
<tr><th>Qn</th><th>Solution</th><th>Comments</th></tr>
<tr><td>iii</td><td>

$\mathrm{E}(S)$

$=\dfrac{6(2)}{(n+5)(n+4)}+\dfrac{7(12)}{(n+5)(n+4)}+\dfrac{8(4n+6)}{(n+5)(n+4)}$
$\quad+\dfrac{9(6n)}{(n+5)(n+4)}+\dfrac{10n(n-1)}{(n+5)(n+4)}$

$=\dfrac{12+84+32n+48+54n+10n^2-10n}{(n+5)(n+4)}$

$=\dfrac{10n^2+76n+144}{(n+5)(n+4)}$

$=\dfrac{(n+4)(10n+36)}{(n+5)(n+4)}$

$=\dfrac{10n+36}{n+5}\text{(Shown)}$

$\mathrm{E}(S^2)$

$=\dfrac{36(2)}{(n+5)(n+4)}+\dfrac{49(12)}{(n+5)(n+4)}+\dfrac{64(4n+6)}{(n+5)(n+4)}$
$\quad+\dfrac{81(6n)}{(n+5)(n+4)}+\dfrac{100n(n-1)}{(n+5)(n+4)}$

$=\dfrac{72+588+256n+384+486n+100n^2-100n}{(n+5)(n+4)}$

$=\dfrac{100n^2+642n+1044}{(n+5)(n+4)}$

$\mathrm{Var}(S)$

$=\mathrm{E}(S^2)-\left[E(S)\right]^2$

$=\dfrac{100n^2+642n+1044}{(n+5)(n+4)}-\dfrac{(10n+36)^2}{(n+5)^2}$

$=\dfrac{(100n^2+642n+1044)(n+5)-(100n^2+720n+1296)(n+4)}{(n+5)^2(n+4)}$

$=\dfrac{100n^3+1142n^2+4254n+5220-(100n^3+1120n^2+4176n+5184)}{(n+5)^2(n+4)}$

$=\dfrac{22n^2+78n+36}{(n+5)^2(n+4)}$

</td><td>For the factorization of $10n^2$ + 76 + 144, use the form given in the question to help. We know that n + 4 should be a factor (so that it cancels with n + 4 in the denominator).
We can use this to check our answer from the previous part as well.</td></tr>
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9 Many electronic devices need a fan to keep them cool. In order to maximise the lifetime of such fans, the speed they run at is reduced when conditions allow. Running a fan at a lower speed reduces the power required. The following table gives details, for a particular type of fan, of the power required (P watts) at different speeds (R revolutions per minute).

Fan speed (R)	3600	4500	5400	6300	7200	8100	9000	9900
Power (P)	0.22	0.34	0.52	0.78	1.06	1.45	2.04	2.64

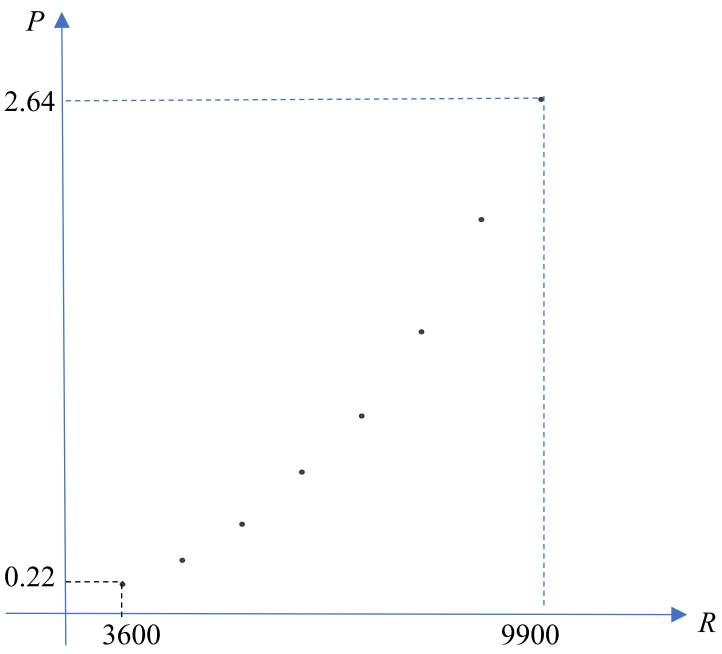
(i) Draw a scatter diagram of these data. Use your diagram to explain whether the relationship between P and R is likely to be well modelled by an equation of the form $P = aR + b$, where a and b are constants. [2]

(ii) By calculating the relevant product moment correlation coefficients, determine whether the relationship between P and R is modelled better by $P = aR + b$ or by $P = aR^2 + b$. Explain how you decide which model is better, and state the equation in this case. [5]

(iii) Use your equation to estimate the speed of the fan when the power is 0.9 watts. Explain whether your estimate is reliable. [2]

(iv) Use your equation to estimate the power used when the speed of the fan is 3300 revolutions per minute. Explain whether your estimate is reliable. [2]

(v) Re-write your equation from part **(ii)** so that it can be used when the speed of the fan, R, is given in revolutions per second. [1]

<table>
<tr><th>Qn</th><th>Solution</th><th>Comments</th></tr>
<tr><td>9i</td><td>

Since the points of the scatter diagram do not appear to lie close to a straight line (P increases by an increasing amount when R increases), the relationship between P and R is not likely to be well modelled by an equation of the form $P = aR + b$.</td><td>For the scatter diagram, the points should be equally spaced in the R-direction.

Can use cross or dots to denote the points, although crosses are preferred.

The values on the axes should also encompass the data range.</td></tr>
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<tr><th>Qn</th><th>Solution</th><th>Comments</th></tr>
<tr><td>ii</td><td>For $P = aR + b$, r-value = 0.969

For $P = aR^2 + b$, r-value = 0.993

The points on the scatter points suggests that as R increases, P increases by increasing amounts, which is consistent with a relationship modelled by $P = aR^2 + b$.
Furthermore, the r-value for $P = aR^2 + b$ is closer to 1, than that of $P = aR + b$.
Hence, $P = aR^2 + b$ is a better model.

$P = 2.8457\times10^{-8}R^2 - 0.28261$
The equation is $P = 2.85\times10^{-8}R^2 - 0.283$.</td><td>Comparing r value is a simple way to explain why particular models are better.

Remember to write the equation in terms of P and R.</td></tr>
<tr><td>iii</td><td>$0.9 = 2.8457\times10^{-8}R^2 - 0.28261$
$R = 6446.5$
≈ 6450 (3 s.f)
When the power is 0.9 watts, the speed of the fan is 6450 revolutions per minute.

The estimation is reliable because 0.9 is within the range of values of P --- $0.22 \le P \le 2.64$ as given in the question, and r-value = 0.993 is very close to 1.</td><td></td></tr>
<tr><td>iv</td><td>$P = 2.8457\times10^{-8}(3300)^2 - 0.28261$
$= 0.027287$
Power used when the speed of the fan is 3300 revolutions per minute is 0.0273 Watts.

The estimation is not reliable because 3300 is not within the range of values ($3600 \le R \le 9900$) of R given in the question, that is extrapolation is used.</td><td></td></tr>
<tr><td>v</td><td>If R is given in revolutions per second,

Method 1:
Replace R with $60R$,
$P = 2.8457\times10^{-8}(60R)^2 - 0.28261$
$P \approx 1.02\times10^{-4}R^2 - 0.283$

Method 2
<table><tr><td>R</td><td>60</td><td>75</td><td>90</td><td>105</td><td>120</td><td>135</td><td>150</td><td>165</td></tr><tr><td>P</td><td>0.22</td><td>0.34</td><td>0.52</td><td>0.78</td><td>1.06</td><td>1.48</td><td>2.04</td><td>2.64</td></tr></table>
By GC,
$P = 1.0245\times10^{-4}R^2 - 0.28261$</td><td>To have a sense of the question,
1 rev per min = $\frac{1}{60}$ rev per minute.
Essentially we want to calculate the new values of R by dividing the original values in the question by 60. This can be seen as a stretch parallel to the R-axis by a factor of $\frac{1}{60}$.
Alternatively, we can just do the actual division by 60 to all the R values and perform the calculation again.</td></tr>
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10 In this question you should state the parameters of any distributions that you use.

A manufacturer produces specialist light bulbs. The masses in grams of one type of light bulb have the normal distribution $\mathrm{N}\left(50, 1.5^2\right)$.

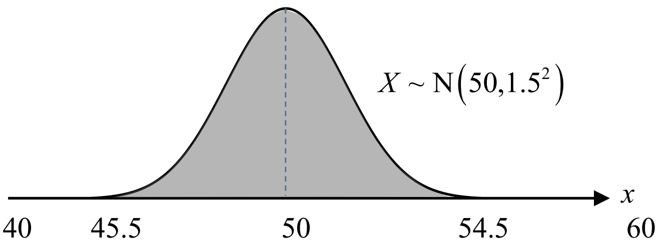
(i) Sketch the distribution for masses between 40 grams and 60 grams. [2]
(ii) Find the probability that the mass of a randomly chosen bulb is less than 50.4 grams. [1]

Each light bulb is packed into a randomly chosen box. The masses of the empty boxes have the distribution $\mathrm{N}\left(75, 2^2\right)$.

(iii) Find the probability that the total mass of 4 randomly chosen empty boxes is more than 297 grams. [2]
(iv) Find the probability that the total mass of a randomly chosen light bulb and a randomly chosen box is between 124.9 and 125.7 grams. [3]

In order to protect the bulbs in transit each bulb is surrounded by padding before being packed in a box. The mass of the padding is modelled as 30% of the mass of the bulb.

(v) The probability that the total mass of a box containing a bulb and padding is more than k grams is 0.9. Find k. [4]
(vi) Find the probability that the total mass of 4 randomly chosen boxes, each containing a bulb and padding, is more than 565 grams. [3]

Qn	Solution	Comments
10i	Let X be the mass in grams of a light bulb of the type of light bulbs produced X~N(50, 1.5^2) 	Remember to state the parameters of any distributions used for every part of this question (**Question wants this!**). From GC, $\mathrm{P}\left(40 < X < 60\right) \approx 1$ so we would need to show that the curve is touch the axis at 40 and 60.
ii	$\mathrm{P}\left(X < 50.4\right) = 0.605$ (3 s.f.)	
iii	Let Y be the mass of an empty box $Y \sim$ N(75, 2^2) Let $S = Y_1 + Y_2 + Y_3 + Y_4$ E(S) = 4(75) = 300, Var(S) = 4(2^2) = 16 $S \sim$ N(300, 16) P(S > 297) = 0.773 (3 s.f.)	$S = Y_1 + Y_2 + Y_3 + Y_4$ not $S = 4Y$
iv	Let $T = X + Y$ E(T) = 50 + 75 = 125, Var(T) = 1.5^2 + 2^2 = 6.25 $T \sim$ N(125, 6.25)	

Qn	Solution	Comments
	$\mathrm{P}\left(124.9 < T < 125.7\right) = 0.126$ (3 s.f.)	
v	Let P g be mass of a padding, A g be the total mass of a padding, box and bulb. $P = 0.3X_2 \sim \mathrm{N}(15, 0.3^2(1.5^2))$ Let $A = X_1 + P + Y$ $\mathrm{E}(A) = 50 + 15 + 75 = 140$, $\mathrm{Var}(A) = 1.5^2 + 0.3^2(1.5^2) + 2^2 = 6.4525$ $A \sim \mathrm{N}(140, 6.4525)$ $\mathrm{P}(A > k) = 0.9$ $k = 136.74 \approx 137$	Note that $A = X_1 + P + Y$ **Not** $A = 1.3X + Y$ If you do not have the newer OS, you will need $\mathrm{P}(A < k) = 0.1$ as an intermediate working.
vi	Let $B = A_1 + A_2 + A_3 + A_4$ $\mathrm{E}(B) = 4(140) = 560$, $\mathrm{Var}(B) = 4(6.4525) = 25.81$ $B \sim \mathrm{N}(560, 25.81)$ $\mathrm{P}(B > 565) = 0.163$	

List of solution videos

Paper 2	URL
1i	https://vimeo.com/366402802
1ii	https://vimeo.com/366402876
2a	https://vimeo.com/366141579
2bi	https://vimeo.com/366141648
2bii	https://vimeo.com/366141691
2biii	https://vimeo.com/366141722
3	https://vimeo.com/366680411
4	https://vimeo.com/364200348/dd2c63db04
5(i)	https://vimeo.com/366139947
5(ii)	https://vimeo.com/366139972
5(iii)	https://vimeo.com/366139989/9e0a0d8e66
6	https://vimeo.com/366643192
7(i)	https://vimeo.com/366146623
7(ii)	https://vimeo.com/366146667/6c96ea60d9
7(iii)	https://vimeo.com/366146693
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9	https://vimeo.com/366139756/efbde43a08
10i	https://vimeo.com/366141741
10iii	https://vimeo.com/366141767
10iv	https://vimeo.com/366141801
10v	https://vimeo.com/366141821
10vi	https://vimeo.com/366141844